



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO RIYA AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

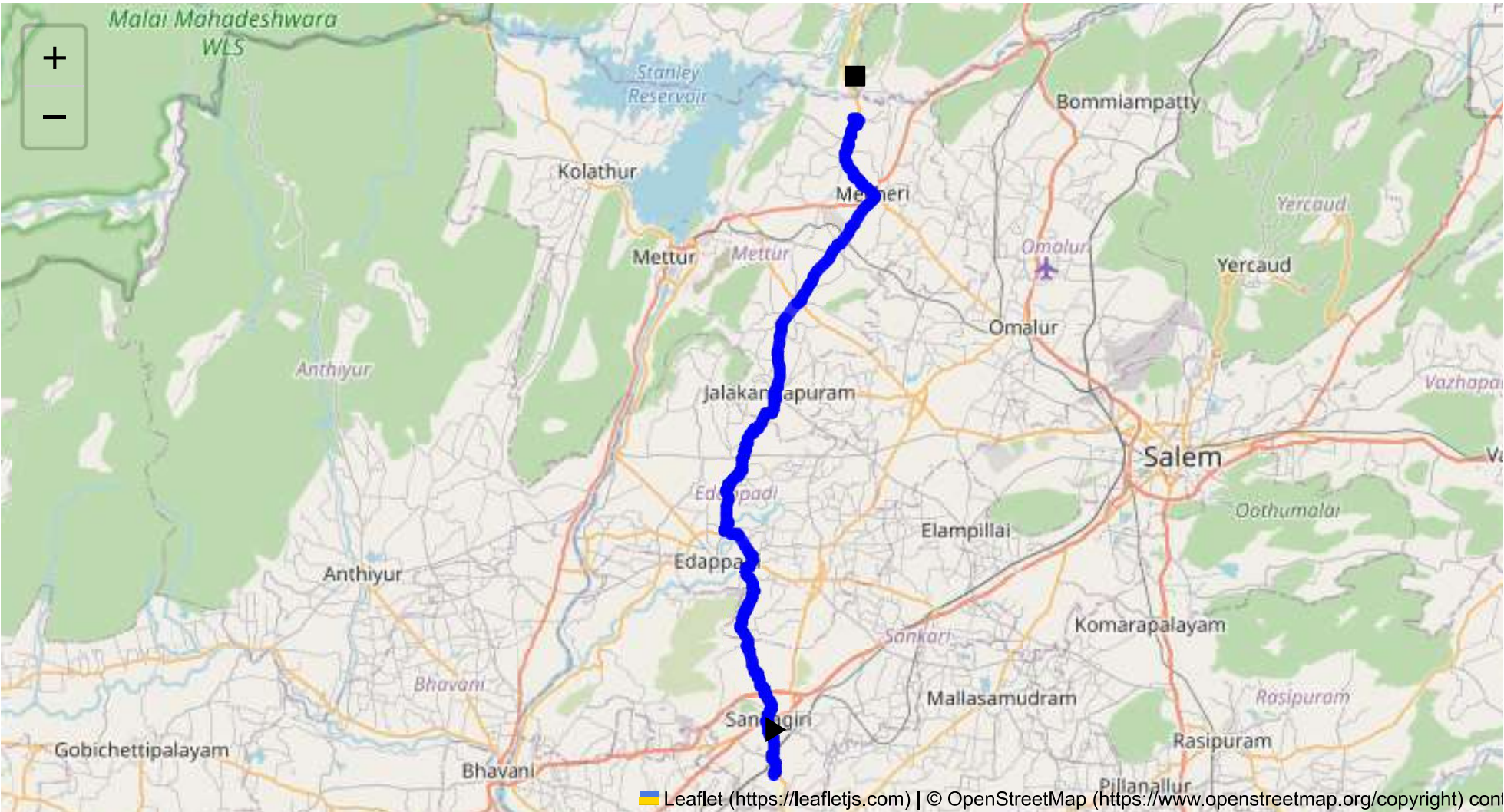
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 58.82 km
Estimated Duration: 1.6 hours
Adjusted Duration (Heavy Vehicle): 2.0 hours
Start: (11.4381, 77.8734)
End: (11.8841129, 77.9285682)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to Mettur covers approximately 58.82 kilometers. Starting at CVQF+23W, Sangagiri, it proceeds via NH44, moving west until branching off towards Mettur. The journey typically transitions from national highways to state roads, leading to semi-urban and rural areas.

2. Typical Weather Conditions and Potential Hazards

The region generally experiences a tropical climate. Weather conditions include:

- **Summer (March to June):** High temperatures can affect road surfaces and vehicle performance.
- **Monsoon (July to September):** Heavy rains can lead to flooding, landslides, and reduced visibility.
- **Winter (October to February):** Mild conditions with sporadic rain; fog may occur in the early mornings affecting visibility.

Weather-related hazards are pronounced during the monsoon due to flooding, waterlogged roads, and reduced traction.

3. Traffic Patterns

- **Peak Hours:** Typically between 8:00 AM - 10:00 AM and 5:00 PM - 7:00 PM.
- **Congestion-Prone Areas:** Sangagiri town center and intersections transitioning from NH44 to narrower roads leading to Mettur.

4. Assessment of Road Quality and Infrastructure

- **National Highway (NH44):** Generally well-maintained, with good signage and lighting.
- **State Roads:** May have uneven surfaces, potholes, and limited shoulder space, especially in rural segments.

5. Suggestions for Alternative Routes

- An alternative route can be the Sangagiri to Erode route via NH544, although longer, it has better infrastructure and fewer rural roads.
- In the event of adverse weather conditions affecting roads, the NH44 towards Salem and further to NH544 towards Mettur can be considered.

6. Local Regulations Affecting Hazardous Material Transport

- Night-time transport of hazardous materials is generally discouraged due to visibility issues.
- Specific permits may be required for transporting certain classes of hazardous materials.

7. Overview of Historical Incidents

- No major historical incidents recorded within the last five years specifically between Sangagiri and Mettur concerning hazardous materials.
- Occasional vehicle breakdowns and minor accidents are more frequent in poorly lit stretches of the route.

8. Environmental Considerations and Sensitive Areas

- Proximity to agricultural land requires awareness to prevent chemical spills.

- Mettur dam vicinity — special attention to vehicular emissions and spillage controls.

9. Analysis of Communication Coverage

- Good cellular network coverage along NH44.
- Possible dead zones: Rural state roads may have inconsistent network coverage.

10. Estimated Emergency Response Times

- **Sangagiri to NH44:** Emergency services can take approximately 15-20 minutes.
- **From transition off NH44 to Mettur:** Response time may extend to 30-40 minutes due to rural road conditions.

11. Summary of Risk Assessment

- The route’s main risks stem from weather conditions, road quality, and potential communication dead zones.
- Traffic during peak hours and road congestion poses additional risks.
- Recommended precautions include weather monitoring, vehicle readiness checks, alternate route planning, and emergency contacts for critical junctures.

Overall, while the route is manageable, it requires clear communication, adherence to local regulations, and vigilance during adverse weather for safe transport of hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60348, 77.83967	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.44900, 77.87406	30 KM/Hr
6	Turn	Medium	11.45000, 77.87213	30 KM/Hr
7	Turn	High	11.45028, 77.87202	15 KM/Hr
8	Turn	Medium	11.45309, 77.87325	30 KM/Hr
9	Turn	Medium	11.46400, 77.87323	30 KM/Hr
10	Turn	Medium	11.46765, 77.87037	30 KM/Hr
11	Turn	High	11.47403, 77.86841	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	High	11.47451, 77.86914	15 KM/Hr
13	Turn	High	11.47920, 77.86989	15 KM/Hr
14	Turn	Medium	11.48070, 77.87166	30 KM/Hr
15	Turn	Medium	11.48155, 77.87174	30 KM/Hr
16	Turn	Medium	11.48277, 77.87250	30 KM/Hr
17	Turn	Medium	11.49496, 77.86660	30 KM/Hr
18	Turn	Medium	11.52571, 77.85522	30 KM/Hr
19	Turn	Medium	11.52772, 77.85585	30 KM/Hr
20	Turn	Medium	11.53857, 77.85087	30 KM/Hr
21	Turn	Medium	11.55027, 77.85506	30 KM/Hr
22	Turn	Medium	11.55460, 77.85624	30 KM/Hr
23	Turn	High	11.57893, 77.85430	15 KM/Hr
24	Turn	Medium	11.57903, 77.85441	30 KM/Hr
25	Turn	Medium	11.58000, 77.85722	30 KM/Hr
26	Turn	Medium	11.60960, 77.84159	30 KM/Hr
27	Turn	Medium	11.61795, 77.84020	30 KM/Hr
28	Turn	Medium	11.62066, 77.84105	30 KM/Hr
29	Turn	Medium	11.64055, 77.84600	30 KM/Hr
30	Turn	Medium	11.64173, 77.84878	30 KM/Hr
31	Turn	Medium	11.64522, 77.85113	30 KM/Hr
32	Turn	Medium	11.65465, 77.85322	30 KM/Hr
33	Turn	Medium	11.66451, 77.85595	30 KM/Hr
34	Turn	Medium	11.67322, 77.86045	30 KM/Hr
35	Turn	Medium	11.67368, 77.86223	30 KM/Hr
36	Turn	Medium	11.68196, 77.86690	30 KM/Hr
37	Turn	High	11.68488, 77.86889	15 KM/Hr
38	Turn	Medium	11.68463, 77.87262	30 KM/Hr
39	Turn	Medium	11.68486, 77.87288	30 KM/Hr
40	Turn	Medium	11.69289, 77.87444	30 KM/Hr
41	Blind Spot	Blind Spot	11.69469, 77.87589	10 KM/Hr
42	Turn	High	11.69491, 77.87546	15 KM/Hr
43	Turn	High	11.69501, 77.87546	15 KM/Hr
44	Turn	High	11.69551, 77.87442	15 KM/Hr
45	Turn	Medium	11.69908, 77.87472	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
46	Turn	Medium	11.69934, 77.87500	30 KM/Hr
47	Turn	Medium	11.76019, 77.89047	30 KM/Hr
48	Turn	Medium	11.75964, 77.89200	30 KM/Hr
49	Blind Spot	Blind Spot	11.75947, 77.89214	10 KM/Hr
50	Blind Spot	Blind Spot	11.83264, 77.94408	10 KM/Hr
51	Turn	Medium	11.84525, 77.92965	30 KM/Hr
52	Turn	Medium	11.85646, 77.92369	30 KM/Hr
53	Turn	Medium	11.88081, 77.92996	30 KM/Hr
54	Turn	Medium	11.88136, 77.93109	30 KM/Hr
55	Blind Spot	Blind Spot	11.88401, 77.93223	10 KM/Hr
56	Turn	Medium	11.88463, 77.92981	30 KM/Hr
57	Turn	Medium	11.88452, 77.92945	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
1	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
7	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	hospital	Parusuraman Hospital	11.697164, 77.877792	30 km/h	Medium
9	hospital	JP Nature Cure Hospital Siddha and Naturopathy	11.7002949, 77.8770897	30 km/h	Medium
10	hospital	SSS Hospital	11.7501471, 77.8832809	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
11	clinic	DSR Clinic	11.7603176, 77.8908853	30 km/h	Medium
12	hospital	Srilakshmi Hospital	11.7601, 77.892272	30 km/h	Medium
13	hospital	Government Hospital	11.758158, 77.894528	30 km/h	Medium
15	hospital	Vasu Speciality Hospital	11.8331097, 77.943169	30 km/h	Medium
16	hospital	Rajamanickam Hospital	11.8332408, 77.9431081	30 km/h	Medium
17	hospital	Child Hospital	11.833001, 77.943415	30 km/h	Medium
18	hospital	Naveen Nursing Home	11.8335355, 77.9437441	30 km/h	Medium
19	clinic	Sri Krishna Clinic	11.835443, 77.9450587	30 km/h	Medium
20	hospital	Sri Sakthi Hospital	11.835714, 77.9451014	30 km/h	Medium
21	hospital	Rishan Hospital	11.879358, 77.9286133	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
6	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium
14	college	Kavery Educational Institutions	11.8255319, 77.9342256	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.58300, 77.85813



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.60348, 77.83967



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544

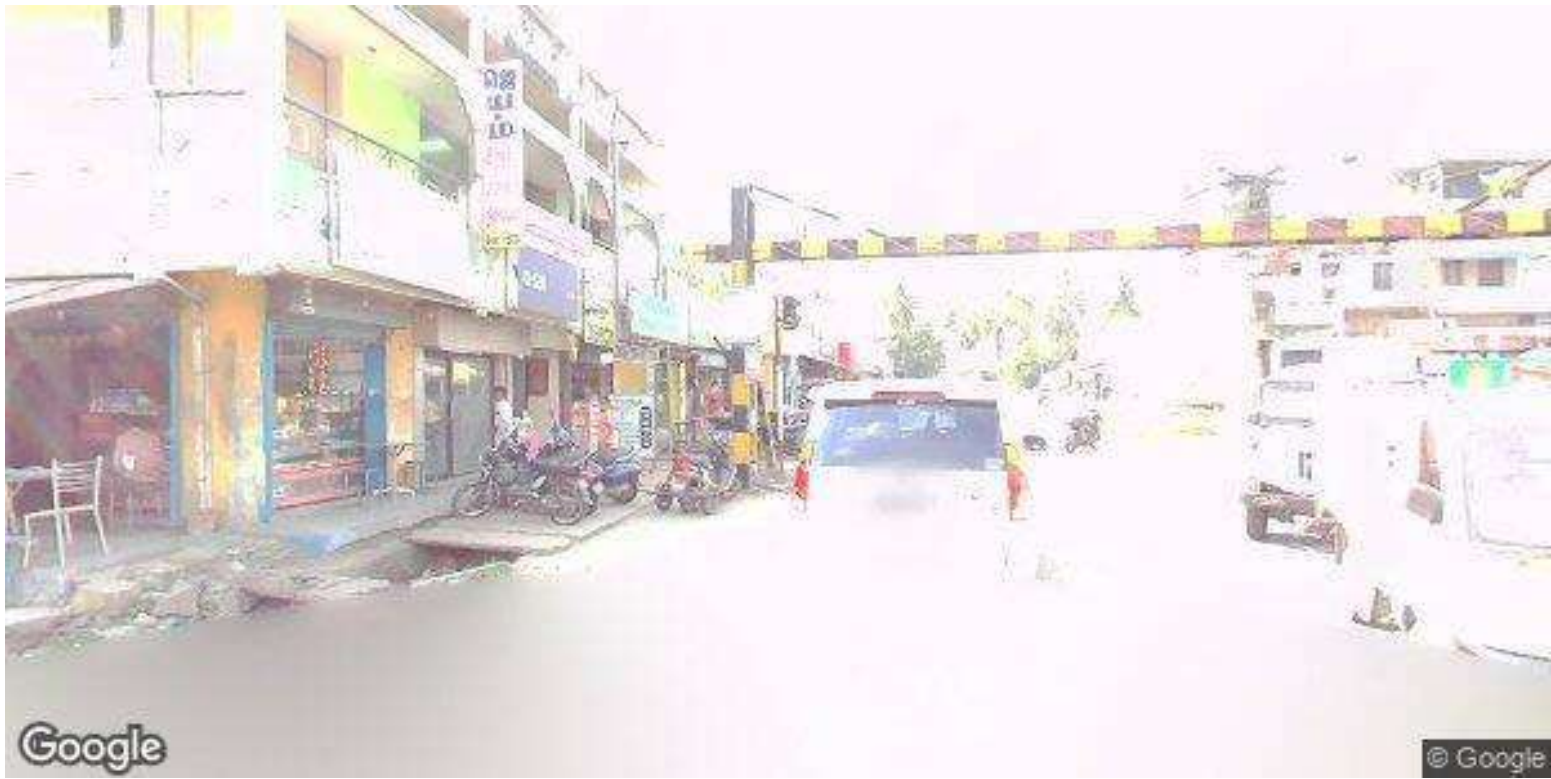


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44900, 77.87406



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45000, 77.87213



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45028, 77.87202



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45309, 77.87325



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46400, 77.87323



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46765, 77.87037



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47403, 77.86841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47451, 77.86914



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47920, 77.86989



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48070, 77.87166



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48155, 77.87174



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48277, 77.87250



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49496, 77.86660



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52571, 77.85522



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.52772, 77.85585



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.53857, 77.85087



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55027, 77.85506



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55460, 77.85624



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57893, 77.85430



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57903, 77.85441



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58000, 77.85722



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.60960, 77.84159



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61795, 77.84020



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.62066, 77.84105



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64055, 77.84600



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64173, 77.84878



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64522, 77.85113



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.65465, 77.85322



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66451, 77.85595



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67322, 77.86045



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67368, 77.86223



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68196, 77.86690



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68488, 77.86889



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68463, 77.87262



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68486, 77.87288



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.69289, 77.87444



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.69469, 77.87589



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69491, 77.87546



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69501, 77.87546



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69551, 77.87442



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69908, 77.87472



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69934, 77.87500



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.76019, 77.89047



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.75964, 77.89200



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.75947, 77.89214



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.83264, 77.94408



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.84525, 77.92965



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.85646, 77.92369



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.88081, 77.92996



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.88136, 77.93109



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.88401, 77.93223